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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
Simple repeats and low-complexity regions were excluded (Cook et al. 2020 Seidl et al. 2020).
Attrition
not required.
Sex as a biological variable
not required.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
1.18.1 package (Gel et al. 2016) and performed 10,000 iterations, using the mean distance to evaluate the closest relationship (bp distance) between SV-breakpoints and TE elements, and circular randomization to maintain the order and distance of the regions in the chromosomes.
Blinding
not detected.
Power Analysis
not detected.
Replication
Single-end sequencing reads of three biological replicates per condition were mapped to V. dahliae strain JR2 genome assembly (Faino et al. 2015) using STAR v.2.4.2.a, allowing multiple mapped reads with the following settings: -outFilterMultimapNmax 100 -winAnchorMultimapNmax

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Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
Briefly, repetitive elements were annotated by using a combination of LTRharvest (Ellinghaus et al. 2008), LTRdigest (Steinbiss et al. 2009), and RepeatModeler.	RepeatModeler		Suggestion: RepeatModeler (RRID:SCR_015027)(link)
The genome-wide occurrence of repeats was determined with RepeatMasker v 4.0.9, and the output was further processed using one code to find them all (Bailly-Bechet et al. 2014).	RepeatMasker		Suggestion: RepeatMasker (RRID:SCR_012954)(link)
Functional gene annotation was based on the available V. dahliae JR2 gene annotation (Faino et al. 2015) and was performed using eggNOG 5.0 (Huerta-Cepas et al. 2019) based on an hierarchical nonsupervised orthology search, restricted to a fungal database with e-values >=0.001 and query coverage >=50.	eggNOG		Suggestion: eggNOG (RRID:SCR_002456)(link)
To define the V. dahliae secretome, we predicted N-terminal signal peptides using SignalP v.	SignalP		
Single-end sequencing reads of three biological replicates per condition were mapped to V. dahliae strain JR2 genome assembly (Faino et al. 2015) using STAR v.2.4.2.a, allowing multiple mapped reads with the following settings: -outFilterMultimapNmax 100 -winAnchorMultimapNmax	STAR		
The bam files were sorted by read name with samtools v.	samtools		
Sequencing reads summarized over TEs and genes were normalized between three replicates in the different conditions using RBioconductor package EdgeR v.	EdgeR		Suggestion: edgeR (RRID:SCR_012802)(link)
Libraries were normalized with TMM method (Robinson and Oshlack 2010) , and converted to CPM-mapped reads using RBioconductor package EdgeR v.	RBioconductor		
Each strain was mapped independently to the reference genome V . dahliae JR2 using BWA -MEM v.	BWA		Suggestion: BWA (RRID:SCR_010910)(link)

Library artifacts were marked and removed using Picard tools v.2.18 with -MarkDuplicates followed by -SortSam to sort the reads (http:broadinstitute.github . iopicard) .	Picard		
Single-nucleotide variants were identified using the -HaplotypeCaller of the Genome Analysis Toolkit (GATK) v.4.0 (Poplin et al. 2018) .	Genome Analysis Toolkit		
	GATK		Suggestion: GATK (RRID:SCR_001876)(link)
Single-nucleotide variants were annotated using SNPeff v.	SNPeff		
Briefly , the workflow takes every strain and maps the genomic reads to the reference genome V . dahliae JR2 using BWA-MEM v.	BWA-MEM		
We used four different callers: DELLY v.	DELLY		
0.8.1 (Rausch et al. 2012) , LUMPY v.2.13 (Layer et al. 2014) , Manta v.1.6.0 (Chen et al. 2016) , and Wham v.1.0 (Kronenberg et al. 2015) in single-sample mode , using each 42 previously sequenced V . dahliae strains independently (supplementary table 1 , Supplementary Material online) .	Wham		
These callers incorporate a diverse range of information for SV detection such as split reads (DELLY , LUMPY , Wham) , discordant read pairs (DELLY , LUMPY , Wham) , read depth signal (LUMPY , Wham) , short-read assembly (Manta , Wham) , and soft-clipping detection (Wham) .	LUMPY		
3.6.3, using packages FactoMineR v.	FactoMineR		
To further investigate the relationship of TEs with gene expression, we summarized TEs in 5 kb windows upstream and downstream genes, using bedtools -window (Quinlan and Hall 2010).	bedtools		
3.6.3 (R Core Team 2019), and visualization with R packages ggplot2 and circlize (Gu et al. 2014 Wickham et al. 2016).	ggplot2		Suggestion: ggplot2 (RRID:SCR_014601)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
To investigate if SVs colocalize , we performed a clustering analysis using CROC (Pignatelli et al. 2009) based on a hypergeometric distribution test and posterior multiple-testing correction using Benjamini-Hochberg CROC scans every chromosome with a sliding window (SV-breakpoints per window = 10 , slide = 1 , >3 SV-breakpoints as a minimum) .	Benjamini-Hochberg
Finally, we computed enrichment association tests using bedtools -fisher v.2.25.0 based on a one-sided Fishers test (Quinlan and Hall 2010) and random association permutation tests using RBioconductor region v.1.18.1 (Gel et al. 2016), performing 5,000 iterations with a resampling randomization.	Fishers test

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq